KAIZEN MATHS

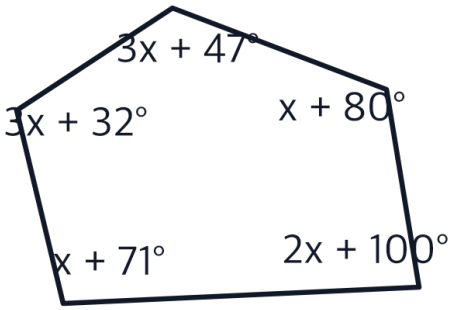
Lesson starter

Polygons and Interior/Exterior Angles

Algebraic Polygons

Name _____

SA1



The interior angles of this pentagon are given in terms of x. Find x.

GEO-BAE7-L3 · SA

Show your working

KAIZEN MATHS

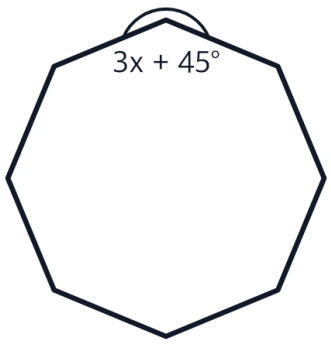
Lesson starter

Polygons and Interior/Exterior Angles

Algebraic Polygons

Name _____

SB1



Each interior angle of a regular octagon is 3x + 45°. Find x.

GEO-BAE7-L3 · SB

Show your working

KAIZEN MATHS

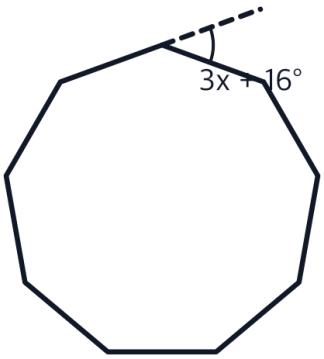
Lesson starter

Polygons and Interior/Exterior Angles

Algebraic Polygons

Name _____

SC1



Each exterior angle of a regular nonagon is 3x + 16°. Find x.

GEO-BAE7-L3 · SC

Show your working

KAIZEN MATHS

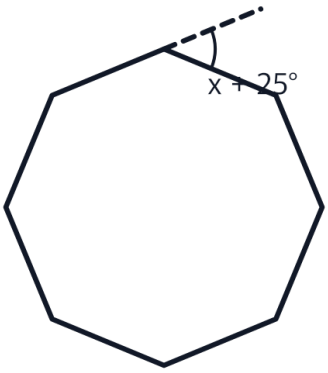
Lesson starter

Polygons and Interior/Exterior Angles

Algebraic Polygons

Name _____

SD1



In a regular polygon, one interior angle is 5x + 35° and the exterior angle at the same vertex is x + 25°. Find x and the number of sides.

GEO-BAE7-L3 · SD

Show your working

KAIZEN MATHS

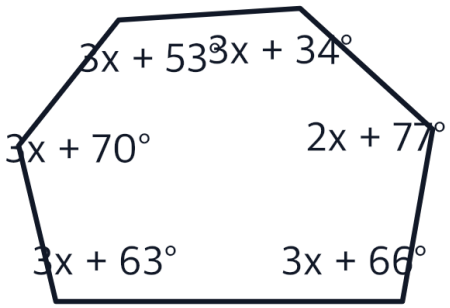
Exit ticket

Polygons and Interior/Exterior Angles

Algebraic Polygons

Name _____

EA1



The interior angles of this hexagon are given in terms of x. Find x.

GEO-BAE7-L3 · EA

Show your working

KAIZEN MATHS

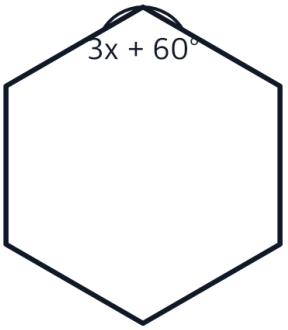
Exit ticket

Polygons and Interior/Exterior Angles

Algebraic Polygons

Name _____

EB1



Each interior angle of a regular hexagon is 3x + 60°. Find x.

GEO-BAE7-L3 · EB

Show your working

KAIZEN MATHS

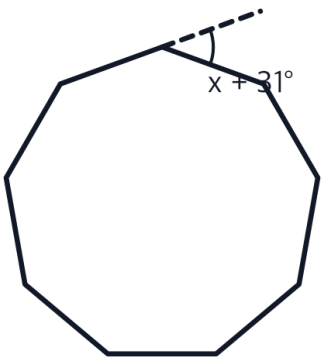
Exit ticket

Polygons and Interior/Exterior Angles

Algebraic Polygons

Name _____

EC1



Each exterior angle of a regular nonagon is x + 31°. Find x.

GEO-BAE7-L3 · EC

Show your working

KAIZEN MATHS

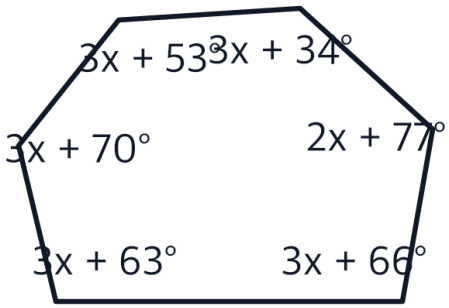
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Polygons and Interior/Exterior Angles

Algebraic Polygons

Name _____

EA1



The interior angles of this hexagon are given in terms of x. Find x.

GEO-BAE7-L3 · EA

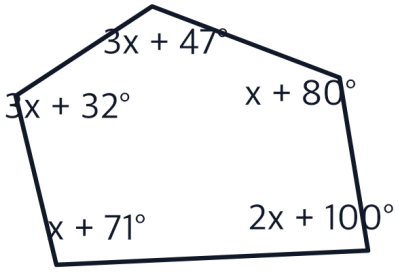
Show your working

Teacher answers & working

Polygons and Interior/Exterior Angles · Algebraic Polygons

GEO-BAE7-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



The interior angles of this pentagon are given in terms of x. Find x.

Answer:

21

Working:

A pentagon has interior angle sum (5 - 2) × 180° = 540°.

Add the given angle expressions and set them equal to 540°.

$(x + 71) + (3x + 32) + (3x + 47) + (x + 80) + (2x + 100) = 540$

$10x + 330 = 540$

$10x = 210$

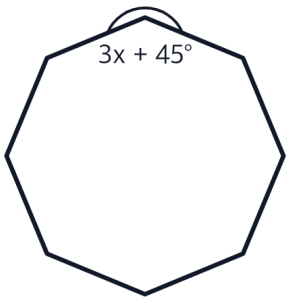
$x = 210 \div 10 = 21.$

Teacher answers & working

Polygons and Interior/Exterior Angles · Algebraic Polygons

GEO-BAE7-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Each interior angle of a regular octagon is 3x + 45°. Find x.

Answer:

30

Working:

First find one interior angle of a regular octagon.

Each exterior angle = 360° ÷ 8 = 45°.

Each interior angle = 180° - 45° = 135°.

Set the expression equal to the interior angle.

$3x + 45 = 135$

$3x = 90$

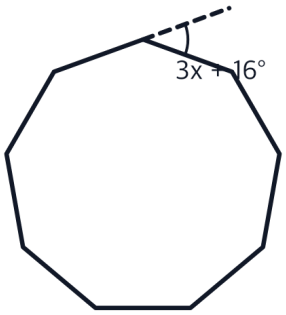
$x = 90 \div 3 = 30.$

Teacher answers & working

Polygons and Interior/Exterior Angles · Algebraic Polygons

GEO-BAE7-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Each exterior angle of a regular nonagon is 3x + 16°. Find x.

Answer:

8

Working:

The exterior angles of a polygon add to 360°.

A regular nonagon has 9 equal exterior angles.

Each exterior angle = 360° ÷ 9 = 40°.

Set the expression equal to the exterior angle.

$3x + 16 = 40$

$3x = 24$

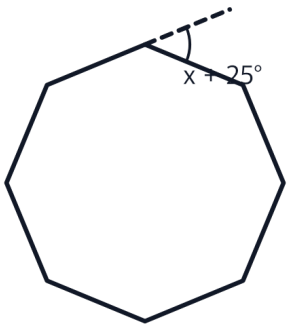
$x = 24 \div 3 = 8.$

Teacher answers & working

Polygons and Interior/Exterior Angles · Algebraic Polygons

GEO-BAE7-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



In a regular polygon, one interior angle is 5x + 35° and the exterior angle at the same vertex is x + 25°. Find x and the number of sides.

Answer:

x = 20, 8 sides

Working:

At one vertex, interior angle + exterior angle = 180°.

$(5x + 35) + (x + 25) = 180$

$6x + 60 = 180$

$6x = 120$

$x = 120 \div 6 = 20.$

Exterior angle = x + 25 = 45°.

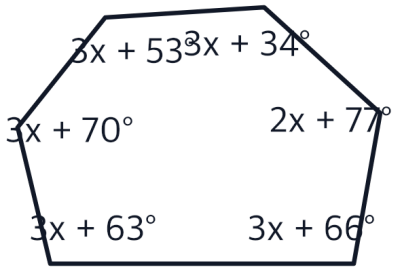
Number of sides = 360° ÷ 45° = 8.

Teacher answers & working

Polygons and Interior/Exterior Angles · Algebraic Polygons

GEO-BAE7-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



The interior angles of this hexagon are given in terms of x. Find x.

Answer:

21

Working:

A hexagon has interior angle sum (6 - 2) × 180° = 720°.

Add the given angle expressions and set them equal to 720°.

$(3x + 63) + (3x + 70) + (3x + 53) + (3x + 34) + (2x + 77) + (3x + 66) = 720$

$17x + 363 = 720$

$17x = 357$

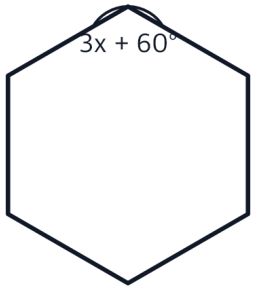
$x = 357 \div 17 = 21.$

Teacher answers & working

Polygons and Interior/Exterior Angles · Algebraic Polygons

GEO-BAE7-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Each interior angle of a regular hexagon is 3x + 60°. Find x.

Answer:

20

Working:

First find one interior angle of a regular hexagon.

Each exterior angle = 360° ÷ 6 = 60°.

Each interior angle = 180° - 60° = 120°.

Set the expression equal to the interior angle.

$3x + 60 = 120$

$3x = 60$

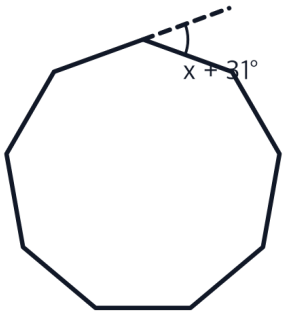
$x = 60 \div 3 = 20.$

Teacher answers & working

Polygons and Interior/Exterior Angles · Algebraic Polygons

GEO-BAE7-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Each exterior angle of a regular nonagon is x + 31°. Find x.

Answer:

9

Working:

The exterior angles of a polygon add to 360°.

A regular nonagon has 9 equal exterior angles.

Each exterior angle = 360° ÷ 9 = 40°.

Set the expression equal to the exterior angle.

$x + 31 = 40$

$x = 9$

$x = 9 \div 1 = 9.$